YOGESH PRADEEP SALVI

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SUMMARY

Robotics and Autonomous Systems MS candidate (May 2026) with hands-on experience in robot manipulation, embedded control, computer vision, and system integration. Proven ability to deliver technical solutions in high uptime environments with measurable performance improvements across hardware and software systems.

EDUCATION

M.S., Robotics and Autonomous Systems | Arizona State University, Tempe, AZ *May 2026*
Coursework: Robotic Systems, Mechatronics, Control Systems, Embedded Systems, Machine Learning, Computer Vision, Sensors & Instrumentation

B.E., Electronics and Telecommunication | Terna Engineering College, Navi Mumbai, India *Spring 2022*
Coursework: Microprocessors, Digital/Analog Electronics, Embedded Systems, Signals and Systems, Control Systems, DSP

TECHNICAL SKILLS

Languages: Python, C, C++, MATLAB
Robotics and Simulation: ROS/ROS2, Gazebo, MuJoCo, Simulink, Rviz, Arduino, Sensors and Actuators
ML and Vision: OpenCV, Machine Learning, Computer Vision, Signal Processing
Design and CAD: SolidWorks, Fusion 360, AutoCAD, 3D Printing

PROFESSIONAL EXPERIENCE

Junior IT and Broadcast Systems Engineer | Pudhari Publications, India *Jun 2023 – Jun 2024*
- To ensure zero-downtime during daily broadcast windows, applied systems-level fault isolation across AV and IT hardware to keep production consistently high.
- Diagnosed hardware and software faults to reduce equipment downtime using root-cause analysis across multi-component systems.
- Standardized AV deployment procedures and technical documentation, improving setup consistency and fault resolution speed.

Systems Integration Intern | Cineom Broadcasting, India *Jul 2022 – Dec 2022*
- Integrated AV systems across live deployments, performing hardware-software co-verification transferable to robotic system bring-up.
- When network and hardware outages threatened live productions, applied structured fault isolation across system layers to identify and resolve root causes quickly.
- Created deployment documentation that cuts setup and troubleshooting time, demonstrating systems-thinking and process discipline.

PROJECTS

Jetson Nano ROS Vision System *Spring 2026*
- Deployed ROS on Jetson Nano with Astra RGB-D camera and OpenCV for real-time object detection at 30 FPS, achieving 95% accuracy.
- Integrated vision system with ROS for dynamic path planning, enabling autonomous navigation with real-time obstacle detection and avoidance while tracking moving targets using costmap updates and local planner adjustments.
- Optimized sensor-to-actuator pipeline through ROS node profiling and refactoring, reducing frame drops by 40% for reliable operation.

6-DOF Cobot Arm Manipulation *Spring 2025*
- Built autonomous maze-solving pipeline using OpenCV visual planning and Python kinematics, executing collision-free paths on a 6-DOF arm.
- Cross-checked FK and IK in MATLAB against the physical arm to confirm consistent end-effector accuracy across the workspace.
- Optimized trajectory interpolation algorithm to reduce actuator jerk by 25%, improving smoothness and extending mechanical component life.

Closed-Loop Path Tracking Robot *Fall 2024*
- Designed PID closed-loop controller in Simulink achieving stable path tracking with real-time encoder feedback on Arduino-based hardware.
- Tuned PID gains through systematic iterative testing to minimize steady-state tracking error across varying speed profiles and surface conditions.
- Validated controller performance by logging and analyzing real-time motor speed and trajectory data across multiple structured test scenarios.

Mechanical Flying Bat *Fall 2024*
- Engineered bio-inspired foldable wing mechanism with ESP32 and IMU feedback for servo-driven actuation simulated adaptive gait in MuJoCo.
- Implemented closed-loop stabilization using IMU sensor fusion to dynamically adjust servo positions and maintain controlled, stable wing actuation.
- Improved glide distance by 35% through iterative wing geometry optimization, balancing aerodynamic lift-to-weight ratio against actuator constraints.

Portable Waveform Generator (Bachelor's Capstone) *Spring 2022*
- Designed ATmega328-based SMD embedded circuit generating stable sine, square, and triangle waveforms using Op-Amp-based signal conditioning.
- Implemented firmware control logic for frequency and amplitude adjustment, enabling configurable waveform output across a defined parameter range.
- Led 3-member team through full hardware build, integration, and field validation cycle, delivering a tested portable instrument for deployment.

OTHER EXPERIENCE

Combat Robot (15 kg): Led 3-member team to design and build a 15 kg combat robot redesigned drivetrain for improved durability and achieved competitive event wins.
Parkinson's Assistive Glove: Built tremor-suppression glove using embedded microcontroller and haptic actuator feedback to reduce involuntary hand motion.
Sun Devil Motorsports (Formula SAE): Performed brake load calculations and fabricated components for ASU's Formula SAE race car.